



Telecom CRM System with Intelligent Automation

Software Requirements Specification (SRS)

V 1.0

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Revision History

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3	V.03	5/5/2026			



1. Introduction

1.1 Background

Telecom companies often face challenges managing customer complaints, subscriptions, and customer retention due to fragmented systems and manual workflows.

These challenges result in:

- Delayed complaint resolution
- Poor customer experience
- Lack of centralized customer information
- High operational workload
- Increased customer churn

This system aims to solve these challenges by introducing a centralized CRM platform integrated with intelligent automation.

1.2 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the **Telecom CRM System with Intelligent Automation**.

This document serves as a reference for stakeholders, developers, testers, business analysts, and project managers to understand the system requirements and expected functionalities.



1.3 Project Summary

The Telecom CRM System with Intelligent Automation is a web-based enterprise CRM platform developed for telecom companies to centralize customer management, complaint handling, package subscriptions, billing overview, and reporting operations.

The system also integrates **n8n workflow automation** to automate repetitive operational tasks such as complaint escalation, package expiry reminders, KPI reporting, and high-priority complaint alerts.

1.4 Intended Audience and Reading Suggestions

This document is intended for:

- Business Analysts
- Project Managers
- Developers
- QA Engineers
- UI/UX Designers
- Stakeholders



1.5 Project Scope

The Telecom CRM System will provide:

- Customer Management
- Complaint Management
- Subscription & Package Management
- Billing Overview
- Notifications System
- Dashboard & Reporting
- Intelligent Workflow Automation (n8n)

Out of Scope

- Mobile customer application
- Payment gateway integration
- Telecom network infrastructure management

1.6 References

- Business Requirements Document (BRD)
- Telecom CRM Business Analysis Documents
- Workflow Automation Documentation (n8n)



2. Overall Description

2.1 Product Perspective

The Telecom CRM System with Intelligent Automation is a standalone web-based application used internally by telecom company employees.

The system centralizes customer operations and integrates workflow automation to improve business efficiency and customer satisfaction.

2.2 Product Features

The system will support the following functions:

- User authentication and authorization
- Customer profile management
- Mobile line management
- Subscription and package management
- Complaint creation and tracking
- Billing overview
- Notifications and alerts
- Dashboard and reporting
- Workflow automation using n8n



2.3 User Classes and Characteristics

Admin

Users responsible for managing system configurations, permissions, and reports.

Customer Service Agent

Users responsible for handling customer interactions, complaints, subscriptions, and billing information.

Supervisor

Users responsible for monitoring complaints, escalation handling, and performance tracking.

Retention Specialist

Users responsible for reducing customer churn and managing retention activities.

2.4 Operating Environment

Web Browsers:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Operating Systems:

- Windows
- macOS
- Internet connection required



2.5 Design and Implementation Constraints

- The project is limited to MVP scope.
- No mobile application in the current phase.
- Payment integration is excluded from MVP.
- Users require authorized access.

2.6 User Documentation

The Telecom CRM System (TeleLine) will include user documentation to support customer service agents, supervisors, and administrators in effectively using the system.

The following documentation will be provided:

1. Customer Service Agent User Guide

A user manual will be provided for customer service agents explaining:

- User login and secure access
- Searching and viewing customer profiles
- Creating complaint tickets
- Updating complaint status
- Viewing customer subscription details
- Renewing or changing telecom packages
- Viewing billing history

2. Supervisor Guide

A dedicated guide will be provided for supervisors covering:

- Complaint assignment
- Complaint escalation management
- SLA monitoring
- KPI dashboard usage
- Complaint reporting and analytics



3. Administrator Guide

An administrator manual will be provided to support:

- User account management
- Role-Based Access Control (RBAC)
- Permission management
- System configuration
- Monitoring n8n automation workflows

4. System Help & Notifications

The system will include built-in guidance such as:

- Error messages
- Validation messages
- Tooltips and field descriptions
- Workflow notifications and alerts

5. Training Materials

Basic onboarding and training materials will be available to help users understand system workflows and business processes.

Documentation Delivery Format

The documentation will be delivered in the following formats:

- PDF User Manuals
- In-System Help Screens
- Internal Training Documents

These documents will ensure proper system usage and improve operational efficiency within telecom customer service operations.



2.7 Assumptions and Dependencies

Assumptions

- Stable internet connection is available.
- Users have required access permissions.
- Customer data can be imported or entered manually.

Dependencies

- n8n automation platform
- SMS notification service
- Email notification service

3. Specific Requirements

3.1 Authentication & Authorization

3.1.1 User Login

- Users should be able to log in using email and password.
- The system should redirect users based on their assigned role.
- The system should support secure authentication.



Validation Rules:

- Email must be in a valid format.
- Password must be correct.
- User account must be active.
- Unauthorized users cannot access the system.

Acceptance Criteria:

- User should successfully log in.
 - User should be redirected to the correct dashboard.
 - Invalid credentials should display an error message.
-

3.1.2 Reset Password

- Users should be able to reset forgotten passwords.
- OTP should be sent to registered email.

Validation Rules:

- Email must already exist in the system.
- OTP must be valid.
- Password must contain:
 - Minimum 8 characters



- One uppercase letter
- One number
- One special character

Acceptance Criteria:

- User can reset password successfully.
- User can log in using the new password.

3.1.3 Role-Based Access Control (RBAC)

- The system should provide access permissions based on user roles.

Roles:

- Admin
- Customer Service Agent
- Supervisor



- Retention Specialist

Acceptance Criteria:

- Users only access authorized modules.
- Restricted pages should not be accessible.

3.2 Customer Management

3.2.1 Add Customer

- Agents should be able to add new telecom customers.
- Required fields:
 - Full Name
 - National ID
 - Phone Number
 - Address



- Mobile Line Number

Validation Rules:

- National ID must be unique.
- Phone number must be valid.
- Required fields cannot be empty.

Acceptance Criteria:

- Customer should be successfully created.
- Customer profile should appear in search results.

3.2.2 Search Customer

- Agents should be able to search customers using:
 - Phone Number
 - National ID
 - Customer Name



Acceptance Criteria:

- Search results should match entered criteria.
 - Customer profile should be accessible.
-

3.2.3 View Customer Profile

- Users should be able to view:
 - Customer Information
 - Mobile Lines
 - Active Package
 - Billing Status
 - Complaint History

Acceptance Criteria:

- Customer data should be displayed accurately.
 - Complaint and billing history should be visible.
-

3.2.4 Edit Customer Information

- Authorized users should update customer information.



Validation Rules:

- Required fields cannot be empty.
- Invalid phone numbers are rejected.

Acceptance Criteria:

- Updated information should be saved successfully.
 - Updated data should be visible immediately.
-

3.3 Complaint Management

3.3.1 Create Complaint Ticket

- Agents should be able to create complaint tickets.

Required Fields:

- Customer
- Complaint Category
- Priority Level
- Description

Complaint Categories:

- Network Issue
- Billing Issue



- Package Issue
- SIM Issue
- Other

Priority Levels:

- Low
- Medium
- High

Validation Rules:

- Required fields cannot be empty.
- Complaint description must contain at least 10 characters.

Acceptance Criteria:

- Complaint should be created successfully.
- Ticket ID should be generated automatically.
- Complaint status should default to "Open".

3.3.2 Complaint Assignment

- Supervisors should assign complaints to agents.



Acceptance Criteria:

- Complaint should be assigned successfully.
 - Assigned agent receives notification.
-

3.3.3 Complaint Status Tracking

Complaint statuses:

- Open
- In Progress
- Escalated
- Resolved
- Closed

Acceptance Criteria:

- Complaint status should update correctly.
 - Complaint timeline should be visible.
-

3.3.4 Complaint Escalation

- Supervisors can escalate unresolved complaints.
- n8n should automatically escalate complaints exceeding SLA.

*Validation Rules:*

- Complaint must exceed SLA period.

Acceptance Criteria:

- Escalated complaint should notify supervisor.
- Complaint priority should update automatically.

3.4 Subscription & Package Management

3.4.1 View Active Package

- Agents should be able to view customer active subscriptions.
- Package details should include:
 - Package Name
 - Package Type
 - Expiration Date
 - Remaining Usage

Acceptance Criteria:

- Active package information should display accurately.
- Package details should update in real time.

3.4.2 Renew Package

- Agents should be able to renew customer subscriptions.

*Validation Rules:*

- Customer account must be active.
- Customer must not have billing restrictions.

Acceptance Criteria:

- Package should renew successfully.
 - Customer receives renewal notification.
-

3.4.3 Upgrade/Downgrade Package

- Agents should be able to upgrade or downgrade customer packages.

Validation Rules:

- Selected package must be available.
- Customer eligibility rules must apply.

Acceptance Criteria:

- Package changes should reflect immediately.
 - Customer receives confirmation notification.
-

3.4.4 Package Expiry Tracking

- System should track package expiration dates.
- The system should notify users before expiration.

Acceptance Criteria:

- Expiry reminders should be sent automatically.
- Expiring packages should appear in dashboard alerts.



3.5 Billing & Payment Overview

3.5.1 View Billing History

- Users should be able to view:
 - Previous Bills
 - Payment Date
 - Payment Amount
 - Due Balance

Acceptance Criteria:

- Billing records should display accurately.
 - Historical billing information should be accessible.
-

3.5.2 Payment Status

Payment Statuses:

- Paid
- Pending
- Overdue

Acceptance Criteria:

- Payment status should display correctly.
 - Outstanding balances should be visible.
-

3.5.3 Outstanding Balance

- Users should view unpaid balances.



Validation Rules:

- Balance must reflect latest payment updates.

Acceptance Criteria:

- Outstanding balance should update correctly.
-

3.6 Dashboard & Reporting

3.6.1 KPI Dashboard

The dashboard should display:

- Total Customers
- Active Complaints
- Resolved Complaints
- Customer Satisfaction Rate
- SLA Compliance
- Revenue Summary

Acceptance Criteria:

- KPI metrics should display accurately.
 - Dashboard should refresh automatically.
-

3.6.2 Complaint Reports

- Supervisors should generate complaint reports.

Reports should include:

- Complaint Category
- Resolution Time



- Complaint Status
- Escalation Rate

Acceptance Criteria:

- Reports should export successfully.
 - Report data should be accurate.
-

3.6.3 Customer Reports

- Admins should generate customer analytics reports.

Reports include:

- Active Customers
- Inactive Customers
- Churn Risk Customers

Acceptance Criteria:

- Reports should generate without errors.
-

3.7 Intelligent Automation (n8n Integration)

3.7.1 Automatic Complaint Escalation

- n8n should automatically escalate complaints exceeding SLA.

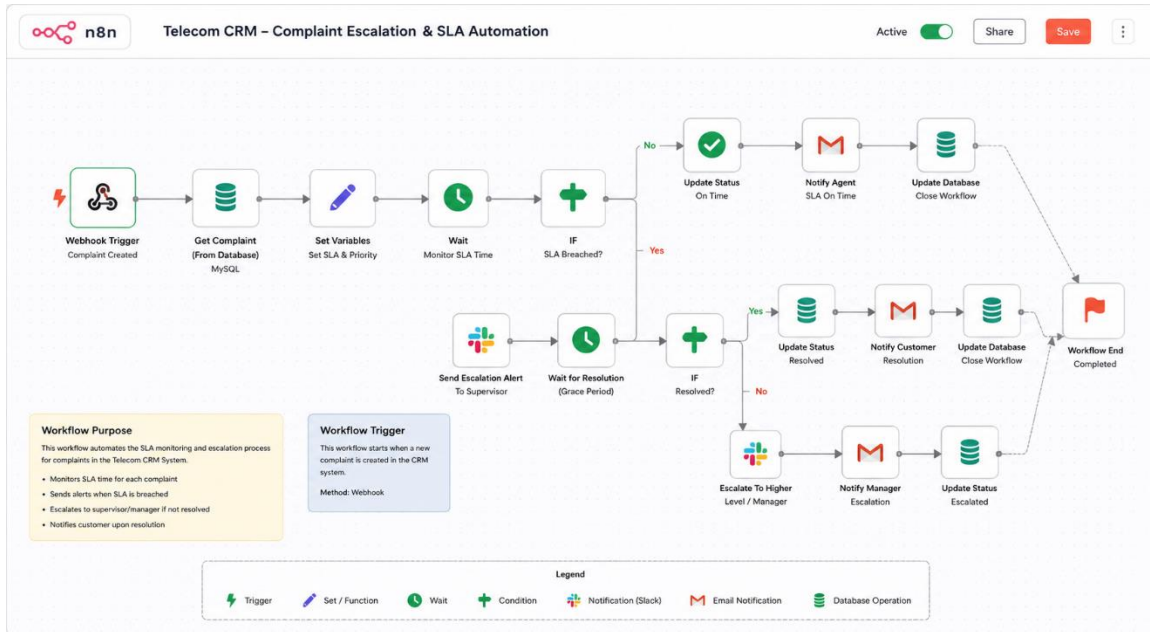
Workflow:

Complaint Created → SLA Timer → Exceed Time → Escalate → Notify Supervisor



Acceptance Criteria:

- Complaint escalates automatically.
- Supervisor receives notification.



3.7.2 Package Expiry Reminder

Description:

The system shall automatically send reminders to customers before package expiration using n8n workflow automation.

Workflow:

Cron Trigger → Check Expiring Packages → IF Expiring < 3 Days → Send SMS / Email Reminder → Update CRM

Functional Requirements:

- The system shall execute a scheduled process daily.
- The system shall identify packages expiring within 3 days.
- The system shall send SMS and Email reminders automatically.
- The system shall update reminder status in CRM.

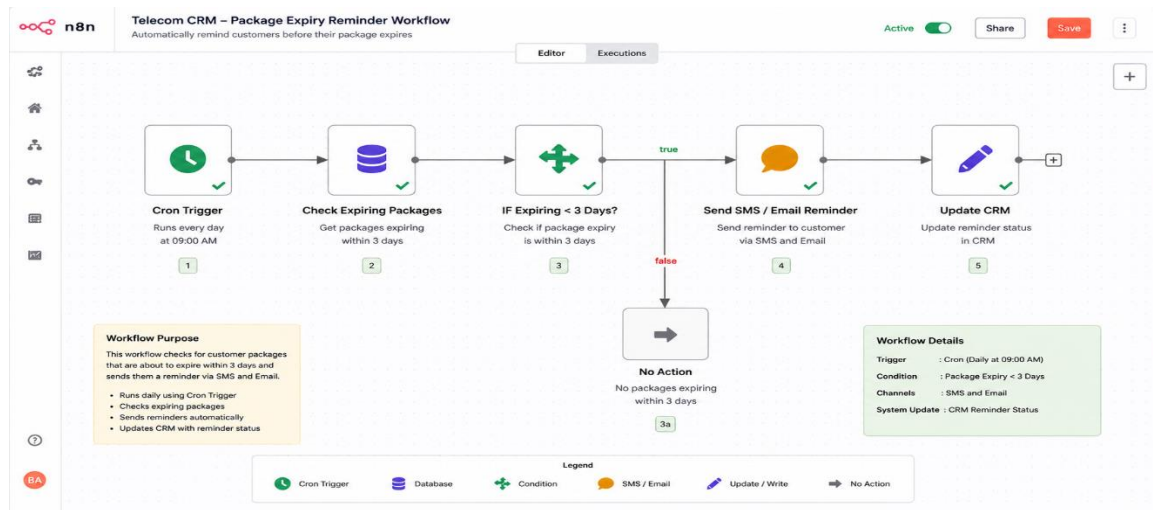


Business Rules:

- Reminders should only be sent to active customers.
- Duplicate reminders should not be sent.
- Expired or inactive packages should be excluded.

Acceptance Criteria:

- Reminder should be sent successfully.
- Customer should receive notification.
- CRM reminder status should be updated



3.7.3 High Priority Complaint Alert

- High-priority complaints should trigger instant alerts.

Acceptance Criteria:

- Assigned team should receive immediate notification.



3.7.4 Weekly KPI Report Automation

Description:

The system shall automatically generate and send weekly KPI reports to management using n8n workflow automation.

Workflow:

Schedule Trigger → Fetch CRM KPIs → Generate Report → Send to Management Email

Functional Requirements:

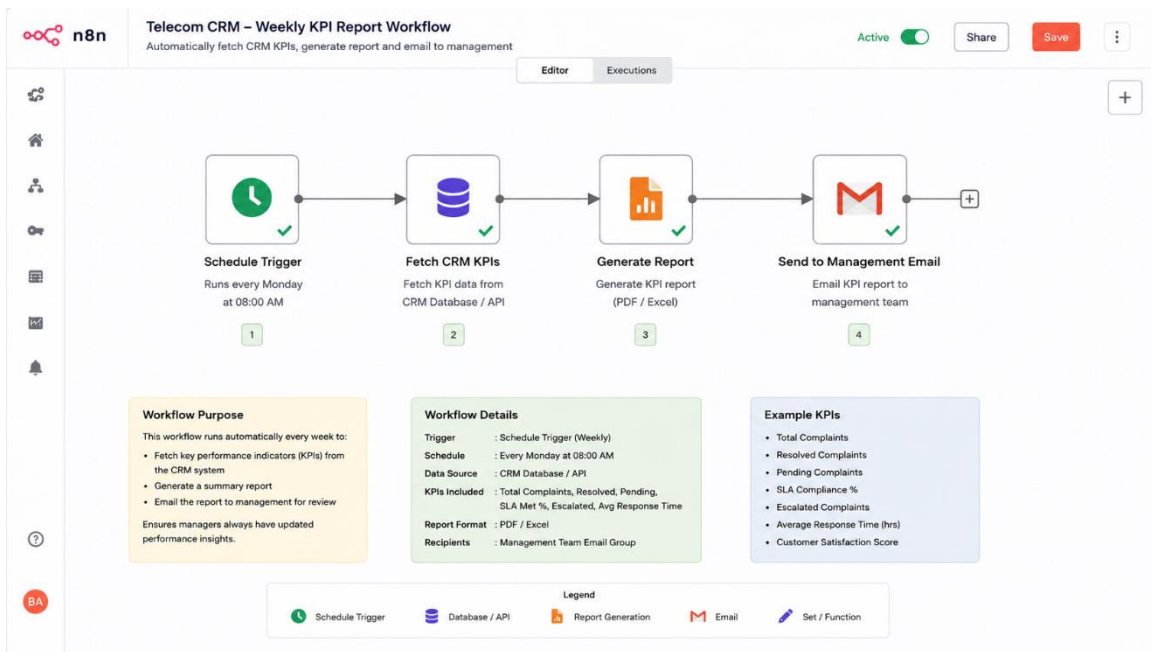
- The system shall generate KPI reports automatically every week.
- The system shall fetch KPI data from CRM database.
- The system shall generate reports in PDF or Excel format.
- The system shall automatically send reports to management email.

KPIs Included:

- Total Complaints
- Resolved Complaints
- Pending Complaints
- SLA Compliance Rate
- Escalated Complaints
- Average Resolution Time
- Customer Satisfaction Score

Acceptance Criteria:

- Weekly report should generate successfully.
- Management should receive report automatically.
- KPI report data should be accurate.





3.8 Use Case Specifications

3.8.1 Use Case UC-001 – User Login

Field	Description
Use Case ID	UC-001
Use Case Name	User Login
Primary Actor	Admin / Customer Service Agent / Supervisor / Retention Specialist
Description	Users log into the Telecom CRM System using valid credentials
Preconditions	User account must exist and be active.
Trigger	User enters login credentials.
Main Flow	1. User enters email and password. 2. System validates credentials. 3. System authenticates user. 4. User redirected to dashboard.
Alternative Flow	Invalid credentials → Error message displayed
Postconditions	User gains authorized system access
Business Rules	Role-Based Access Control (RBAC) applies.
Acceptance Criteria	User logs in successfully and accesses authorized modules.





3.8.2 Use Case UC-002 – Search Customer

Field	Description
Use Case ID	UC-002
Use Case Name	Search Customer
Primary Actor	Customer Service Agent
Description	Agent searches for a telecom customer using available identifiers.
Preconditions	Agent must be logged in.
Trigger	Agent initiates customer search.
Main Flow	1. Agent enters National ID / Phone Number / Customer Name. 2. System searches customer database. 3. Matching customer profile displayed.
Alternative Flow	No customer found → System displays error message.
Postconditions	Customer profile becomes accessible.
Business Rules	Search should support partial matching
Acceptance Criteria	Correct customer information displayed



3.8.3 Use Case UC-003 – Create Complaint Ticket

Field	Description
Use Case ID	UC-003
Use Case Name	Create Complaint Ticket
Primary Actor	Customer Service Agent
Description	Agent creates a complaint ticket for a telecom customer.
Preconditions	Customer profile must exist.
Trigger	Customer submits complaint.
Main Flow	1. Agent searches customer. 2. Agent selects complaint category. 3. Agent enters complaint details. 4. System generates Ticket ID. 5. Complaint saved successfully.
Alternative Flow	No customer found → System displays error message.
Postconditions	Complaint ticket created successfully.
Business Rules	Ticket Priority = Low / Medium / High.
Acceptance Criteria	Complaint created and status defaults to “Open”



3.8.4 Use Case UC-004 – Complaint Escalation

Field	Description
Use Case ID	UC-004
Use Case Name	Complaint Escalation
Primary Actor	Supervisor / n8n Automation
Description	Complaint escalates automatically if SLA time exceeded.
Preconditions	Complaint must remain unresolved beyond SLA
Trigger	SLA timer exceeded.
Main Flow	1. n8n detects SLA breach. 2. System escalates complaint. 3. Supervisor receives notification.
Alternative Flow	Complaint resolved before SLA → No escalation.
Postconditions	Complaint status becomes “Escalated”.
Business Rules	High-priority complaints escalate immediately.
Acceptance Criteria	Supervisor notified successfully



3.8.5 Use Case UC-005 – Renew Package

Field	Description
Use Case ID	UC-005
Use Case Name	Renew Package
Primary Actor	Customer Service Agent
Description	Agent renews customer telecom package
Preconditions	Customer account active
Trigger	Customer requests renewal.
Main Flow	1. Agent opens customer profile. 2. Agent selects renewal option. 3. System validates eligibility. 4. Package renewed.
Alternative Flow	Billing issue → Renewal rejected
Postconditions	Package status updated.
Business Rules	Only active accounts can renew.
Acceptance Criteria	Renewal confirmation displayed.



3.8.6 Use Case UC-006 – View Customer Profile

Field	Description
Use Case ID	UC-006
Use Case Name	View Customer Profile
Primary Actor	Customer Service Agent / Supervisor
Description	Users view customer information, active subscriptions, billing, and complaint history.
Preconditions	Customer must exist in the system.
Trigger	User selects customer profile.
Main Flow	1. User searches customer. 2. System retrieves profile data. 3. Customer information displayed.
Alternative Flow	Customer not found → Error message displayed.
Postconditions	Customer details become accessible.
Business Rules	Only authorized users can access customer profile.
Acceptance Criteria	Profile data displays accurately.



3.8.7 Use Case UC-007 – Assign Complaint

Field	Description
Use Case ID	UC-007
Use Case Name	Assign Complaint
Primary Actor	Supervisor
Description	Supervisor assigns complaint ticket to available agent.
Preconditions	Complaint must exist and status = Open.
Trigger	Supervisor opens complaint queue.
Main Flow	1. Supervisor selects complaint. 2. Supervisor selects available agent. 3. System updates assignment. 4. Agent receives notification.
Alternative Flow	No available agent → Queue remains pending.
Postconditions	Complaint assigned successfully.
Business Rules	High-priority complaints should be assigned first.
Acceptance Criteria	Assigned agent receives notification.



3.8.8 Use Case UC-008 – Update Complaint Status

Field	Description
Use Case ID	UC-008
Use Case Name	Update Complaint Status
Primary Actor	Customer Service Agent / Supervisor
Description	Users update complaint progress.
Preconditions	Complaint must exist.
Trigger	Agent updates ticket progress.
Main Flow	1. User opens complaint. 2. User selects status. 3. System saves update.
Alternative Flow	Invalid update → Validation message.
Postconditions	Complaint status updated.
Business Rules	Allowed statuses: Open, In Progress, Escalated, Resolved, Closed.
Acceptance Criteria	Complaint timeline updated correctly.



3.8.9 Use Case UC-009 – Upgrade/Downgrade Package

Field	Description
Use Case ID	UC-009
Use Case Name	Upgrade/Downgrade Package
Primary Actor	Customer Service Agent
Description	Agent changes customer subscription package.
Preconditions	Customer account active.
Trigger	Customer requests package change.
Main Flow	1. Agent opens package details. 2. Agent selects new package. 3. System validates eligibility. 4. Package updated.
Alternative Flow	Package unavailable → Error displayed.
Postconditions	Subscription updated successfully.
Business Rules	Eligibility rules apply.
Acceptance Criteria	Package changes reflect immediately.



3.8.10 Use Case UC-010 – View Billing History

Field	Description
Use Case ID	UC-010
Use Case Name	View Billing History
Primary Actor	Customer Service Agent / Supervisor
Description	Users view customer billing history and due balance.
Preconditions	Customer profile exists.
Trigger	User selects billing section.
Main Flow	1. User opens billing tab. 2. System retrieves billing history. 3. Payment records displayed.
Alternative Flow	No billing data → Empty state shown.
Postconditions	Billing history visible.
Business Rules	Billing data must be real-time.
Acceptance Criteria	Accurate billing details displayed.



3.8.11 Use Case UC-011 – Generate Complaint Report

Field	Description
Use Case ID	UC-011
Use Case Name	Generate Complaint Report
Primary Actor	Supervisor
Description	Supervisor generates complaint performance report.
Preconditions	Complaint data exists.
Trigger	SLA timer exceeded.
Main Flow	1. Supervisor selects report filters. 2. System fetches complaint KPIs. 3. Report generated. 4. Export/download available.
Alternative Flow	No data available → Empty report displayed.
Postconditions	Report generated successfully.
Business Rules	Reports support export formats (PDF/Excel).
Acceptance Criteria	Report generated without errors.



3.8.12 Use Case UC-012 – Package Expiry Reminder (n8n)

Field	Description
Use Case ID	UC-012
Use Case Name	Package Expiry Reminder
Primary Actor	n8n Automation
Description	n8n automatically sends package expiry reminders to customers.
Preconditions	Package expires within 3 days.
Trigger	Daily scheduled cron trigger.
Main Flow	1. n8n checks expiring packages. 2. System validates active subscriptions. 3. SMS/Email reminder sent. 4. CRM updated.
Alternative Flow	Customer inactive → Reminder skipped.
Postconditions	Reminder sent successfully.
Business Rules	No duplicate reminders.
Acceptance Criteria	Customer notified before expiry.



3.8.13 Use Case UC-013 – Weekly KPI Report Automation (n8n)

Field	Description
Use Case ID	UC-013
Use Case Name	Weekly KPI Report Automation
Primary Actor	n8n Automation
Description	n8n automatically generates KPI reports and emails management.
Preconditions	Weekly schedule active.
Trigger	Weekly scheduler execution.
Main Flow	1. n8n fetches CRM KPIs. 2. System generates report. 3. Report emailed to management.
Alternative Flow	KPI unavailable → Log error.
Postconditions	Weekly report delivered.
Business Rules	Reports generated weekly.
Acceptance Criteria	Management receives accurate KPI report.



3.8.14 Use Case UC-014 – Weekly KPI Report Automation (n8n)

Field	Description
Use Case ID	UC-014
Use Case Name	High Priority Complaint Alert
Primary Actor	n8n Automation
Description	System instantly alerts responsible teams about high-priority complaints.
Preconditions	Complaint priority = High.
Trigger	High-priority complaint created.
Main Flow	1. Complaint created. 2. n8n detects high priority. 3. Alert sent to assigned team.
Alternative Flow	Team unavailable → Escalation occurs.
Postconditions	Team notified immediately.
Business Rules	High-priority alerts must be instant.
Acceptance Criteria	Team receives alert immediately.



3.9 User Stories – Telecom CRM System with Intelligent Automation

Module 1: Authentication & Access Management

US-001 – User Login

As a Customer Service Agent,
I want to securely log into the CRM system,
so that I can access customer information and manage complaints.

Acceptance Criteria:

- User enters valid email and password.
 - System authenticates credentials.
 - User is redirected to dashboard.
 - Unauthorized users cannot access the system.
-

US-002 – Role-Based Access

As an Admin,
I want different users to have role-based permissions,
so that only authorized users can access sensitive features.

Acceptance Criteria:

- Agent can access complaints only.
- Supervisor can assign/escalate complaints.
- Admin has full access.



Module 2: Customer Management

US-003 – Search Customer

As a Customer Service Agent,
I want to search for customers by phone number, national ID, or
name,
so that I can quickly access customer information.

Acceptance Criteria:

- Search results display accurate customer data.
- Partial search is supported.
- Error message shown if customer not found.

US-004 – View Customer Profile

As a Customer Service Agent,
I want to view customer profile details,
so that I can understand subscriptions, billing, and complaint history.

Acceptance Criteria:

- Customer profile loads successfully.
- Active packages displayed.
- Billing history visible.
- Complaint history available.



US-005 – Edit Customer Information

As a Customer Service Agent,
I want to update customer information,
so that customer records remain accurate.

Acceptance Criteria:

- Required fields validated.
- Changes saved successfully.

Module 3: Complaint Management

US-006 – Create Complaint Ticket

As a Customer Service Agent,
I want to create complaint tickets,
so that customer issues can be tracked and resolved.

Acceptance Criteria:

- Ticket ID generated automatically.
 - Complaint category selected.
 - Priority level assigned.
 - Status defaults to “Open”.
-



US-007 – Assign Complaint

As a Supervisor,
I want to assign complaints to agents,
so that issues can be resolved efficiently.

Acceptance Criteria:

- Available agents displayed.
 - Complaint assigned successfully.
 - Assigned agent notified.
-

US-008 – Update Complaint Status

As a Customer Service Agent,
I want to update complaint status,
so that issue progress can be tracked.

Acceptance Criteria:

- Status updates successfully.
 - Timeline reflects changes.
 - Valid statuses only allowed.
-



US-009 – Complaint Escalation

As a Supervisor,

I want unresolved complaints to be escalated automatically,
so that SLA violations are minimized.

Acceptance Criteria:

- SLA breach triggers escalation.
 - Supervisor receives notification.
 - Priority updated automatically.
-

Module 4: Subscription & Package Management

US-010 – View Active Package

As a Customer Service Agent,

I want to view active customer packages,
so that I can support subscription inquiries.

Acceptance Criteria:

- Package details displayed.
 - Expiration date visible.
 - Remaining usage shown.
-



US-011 – Renew Package

As a Customer Service Agent,
I want to renew customer packages,
so that subscriptions remain active.

Acceptance Criteria:

- Renewal processed successfully.
 - Customer receives confirmation.
 - CRM updates package details.
-

US-012 – Upgrade/Downgrade Package

As a Customer Service Agent,
I want to change customer subscription plans,
so that customers receive suitable services.

Acceptance Criteria:

- Eligible packages shown.
 - Changes reflected immediately.
 - Confirmation sent.
-



Module 5: Billing & Payments

US-013 – View Billing History

As a Customer Service Agent,
I want to view customer billing history,
so that I can assist with payment inquiries.

Acceptance Criteria:

- Bills displayed accurately.
 - Payment dates visible.
 - Outstanding balances shown.
-

Module 6: Dashboard & Reporting

US-014 – View KPI Dashboard

As a Supervisor,
I want to view KPI metrics,
so that I can monitor operational performance.

Acceptance Criteria:

- Dashboard refreshes automatically.
 - SLA compliance visible.
 - Complaint statistics accurate.
-



US-015 – Generate Complaint Report

As a Supervisor,
I want to generate complaint reports,
so that I can analyze service performance.

Acceptance Criteria:

- Reports exportable to PDF/Excel.
 - Accurate KPI data displayed.
 - Filters supported.
-

Module 7: Intelligent Automation (n8n)

US-016 – Package Expiry Reminder

As an n8n Automation Workflow,
I want to automatically send package expiry reminders,
so that customers renew before expiration.

Acceptance Criteria:

- Reminder sent 3 days before expiry.
 - SMS/Email delivered successfully.
 - CRM updated.
-



US-017 – Weekly KPI Report Automation

As an n8n Automation Workflow,
I want to automatically generate weekly KPI reports,
so that management receives performance insights.

Acceptance Criteria:

- Weekly schedule runs automatically.
 - KPI report generated.
 - Report emailed successfully.
-

US-018 – High Priority Complaint Alert

As an n8n Automation Workflow,
I want to instantly notify teams for high-priority complaints,
so that urgent cases are handled quickly.

Acceptance Criteria:

- Instant alert triggered.
 - Responsible team notified.
 - Escalation logged.
-



4. Non-Functional Requirements

4.1 Performance Requirements

- System should support up to 10,000 concurrent users.
 - Page load time should be less than 3 seconds.
 - Dashboard reports should load within 5 seconds.
-

4.2 Security Requirements

- Role-Based Access Control (RBAC) must be implemented.
 - Sensitive data must be encrypted.
 - Secure login authentication is required.
 - Unauthorized access should be blocked.
-

4.3 Usability Requirements

- The interface should be intuitive and easy to use.
 - The dashboard should support responsive web design.
 - Navigation should be simple and consistent.
-

4.4 Reliability Requirements

- System availability should be at least 99.9%.
- Automatic backups should be performed regularly.
- System failures should be logged.



4.5 Scalability Requirements

- The system should support future integrations.
- The system should scale to support increasing telecom customers.

5. Appendices

5.1 Assumptions

- Users have access to internet connection.
- SMS and Email services are available.
- Customer data exists or can be migrated.

5.2 Glossary

CRM: Customer Relationship Management

KPI: Key Performance Indicator

SLA: Service Level Agreement

RBAC: Role-Based Access Control

n8n: Workflow Automation Platform

6. Appendix

6.1 Wireframe



A Web Page

← → ↻ https://



Logo

Welcome

[Forget Password ?](#)

[Don't Have an Account ?](#)

A Web Page

← → ↻ https://

TeleLink CRM

Admin Console

Home
Customers
Compliant
Subscription
Reports
Setting

Lead Admin& P

Total Cust

Activ Compl

Activ Compl

Resolu Rate

Compliant Trends (30 Days)

Revenue By Service

Critical Support Ticket

Customer	Issus	Service	Sal Statues	Action

Alert & Naticati



A Web Page

← → ↻ https://

TeleLink CRM

Admin Console

Home

Customers

Compliant

Subscription

Reports

Setting

search Customers , tickets , reports

Ahmed Samy Active VIP

Edit Profil Add Servi

Alix Member since 2023

Customer Summary

Billing Summary

Active Lines 2 Devices

Subscribing Details

Complaint History

A Web Page

← → ↻ https://

TeleLink CRM

Admin Console

Home

Customers

Compliant

Subscription

Reports

Setting

search Customers , tickets , reports

Complaints Queue

Create Complaint

Manage and resolve customer service tickets in real time

Active Ticket

Pending Tech

Avg.Resolution Time

CAST Score

Filter By: All categories All priorities Clear All Filters

Ticket ID	Customer	Category	Priority	Statues	Action



A Web Page

← → ↻ https://

TeleLink CRM

Admin Console

Home
Customers
Compliant
Subscription
Reports
Setting

search Customers , tickets , reports

Subscription Planes

Manage , modify and create telecommunication service

Create New Plane

Total Active Subscription

Expiring Soon

New Sign-up (30 Day's)

All categories ▾ All Statues ▾

Plan Name	Category	Monthly Price	Subscribers	Statues

A Web Page

← → ↻ https://

TeleLink CRM

Admin Console

Home
Customers
Compliant
Subscription
Reports
Setting

search Customers , tickets , reports

Account Setting

Manage Your Personal information , security performance , and system configurations

Profile Security Notifications System Performance

Personal Information

Profile Photo
Upload A new Photo (PNG ,JPG , max 5 MB) This will be visible to you

Upload new Pic Remove

Full Name
Ahmed Samy

Corporate Email
Ahmed.Samy@telelin.corp

Job Title
Lead Admin

Department
Enterprise Operations ▾

Save Changes

Account Summary

Account Type Enterprise Admin

Member Since Jan,12,2022

Last Login Today , 09:42 AM

Account Summary

Change Password >

Active Session >

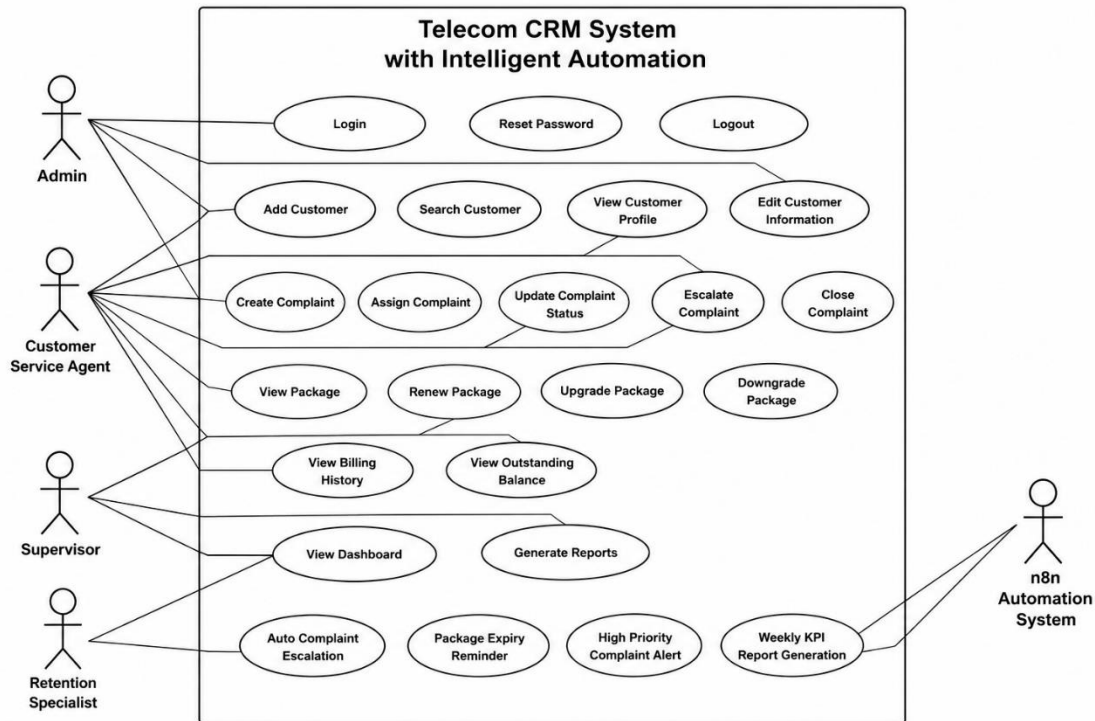
Delete Account

Help
Log-out

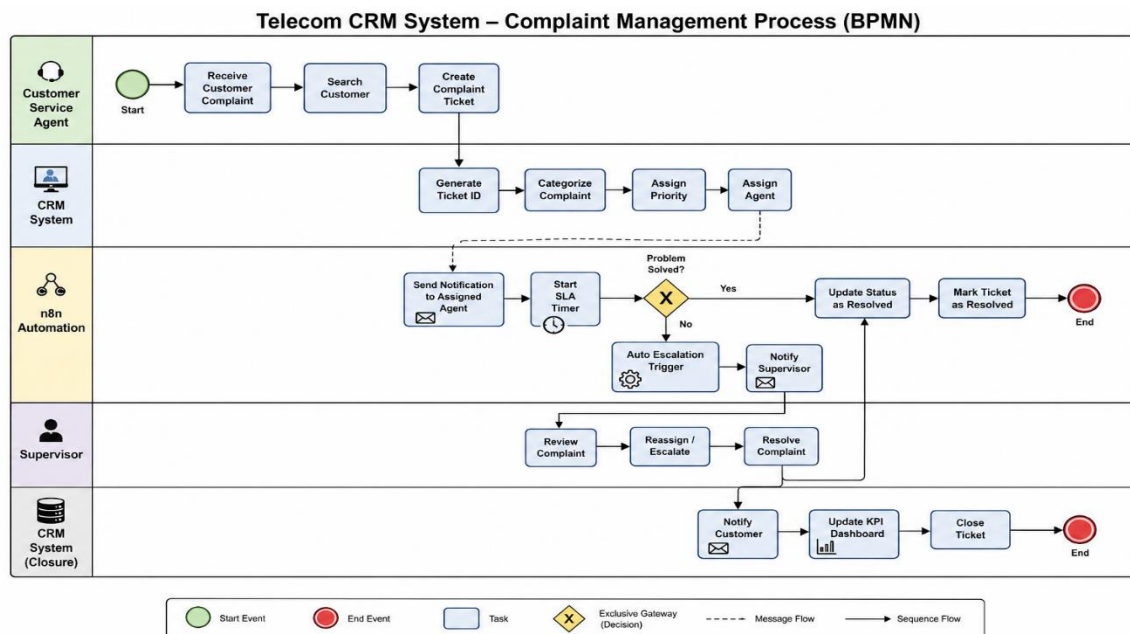


6.2 UML Diagrams

6.2.1 Use case Diagram

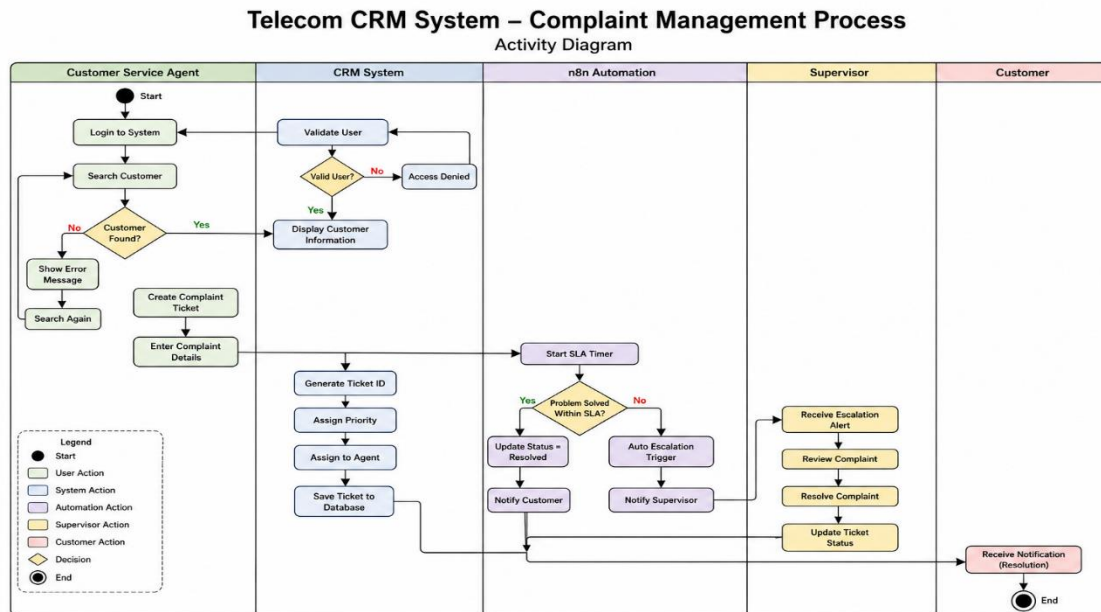


6.2.2 BPMN





6.2.3 Activity Diagram



6.2.4

